CFLMUN'26

Addressing the Usage of Advanced Tech and Chemical Weapons in Ongoing and Recent Histories Armed Conflicts

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STUDY GUIDE

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1. LETTER FROM THE SECRETARY GENERAL

Dear Delegates,

Welcome to CFLMUN'26 everyone! My name is Sümeyra, and I have the honour of serving as the Secretary General at this conference. I'm filled with excitement and happiness. Also, I'd like to thank my Head of Academy Belinay Takcı, my Under Secretary General Bahar Erdoğan and my academic assistant Zeynep Işık for helping to organize the committee.

In the Disarmament and International Security Committee, DISEC, it brings disarmament, which is one of the biggest problems of today, to the age of artificial intelligence and creates an environment of discussion on this subject. I am sure you will touch on very important issues in the committee.

Also, I want to say thank each and every one of you for choosing to be a part of this committee. Your presence and participation are what will make it truly meaningful. I look forward to seeing you all between May 1st and 3rd, ready to challenge perspectives, defend positions, and most importantly, enjoy the experience.

Wishing you all the best of luck. *Let's shape the future in the light of diplomacy!*

Warm regards,

Sümeyra Gülsüm Çolak

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2. LETTER FROM THE HEAD OF ACADEMY

Distinguished delegates,

It is a great honour to welcome you to the first edition of CFLMUN. Over the course of this conference, you will engage in thoughtful debate, share perspectives, and develop solutions to complex global issues. In a world shaped by conflict and uncertainty, your voice carries weight. The responsibility to think critically and act thoughtfully belongs to you. As delegates, you are not only participants in debate, but contributors to ideas that shape perspectives beyond this room.

Your dedication to preparing for this conference already shows your commitment to building a better future. Use this opportunity to challenge your critical thinking, refine your voice, and explore perspectives different from your own. We look forward to seeing the creativity, determination, and insight you bring to your committees.

It's essential not to forget that the experience you acquire here will stay with you for years to come. Make the most of every moment, both inside and outside your committee sessions. Do not hesitate to seek guidance whenever you need it.

Your seat is ready. We are eager to hear your ideas.

Sincerely,

Belinay Takcı

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3. LETTER FROM THE ACADEMIC ASSISTANT OF DISEC

Distinguished Delegates,

I sincerely welcome you all to our committee. I believe that our agenda item is one of the most challenging and significant topics within the United Nations. Although it may seem intimidating at first, I have full confidence in each and every one of you. You must be brave enough to confront the harsh realities of our world, especially when you have both the opportunity and the potential to turn these challenges into meaningful solutions.

Over the next three days, you will be, in many ways, the individuals shaping the course of this committee. I encourage you to use this opportunity to create impactful and lasting change. As selected delegates, we expect you to be open-minded, determined, and courageous. Speak clearly, debate effectively, and most importantly, be the change you wish to see.

At times, your words may feel small; however, never forget that meaningful change begins with dialogue. You have the power to contribute to a better future, a more just society, and a more livable world. Ensure that your solutions are realistic, sustainable, and inclusive—because true change must benefit everyone.

Please make sure to read this study guide carefully. Everything you will need for the upcoming three days is provided within it. Pay close attention to the details, as they may inspire your solutions. While the main topics are outlined at the end of this guide, feel free to explore beyond them, as our time allows for deeper discussion.

By the end of this conference, I hope you will not only gain awareness of the harsh realities we often overlook, but also develop the determination to address them. *Let this conference be a space for awareness, meaningful debate, and effective solutions.*

Sincerely,

Zeynep Işık

Academic Assistant of DISEC

4. INTRODUCTION TO THE COMMITTEE

a. What Is DISEC?

After the UN Charter was signed in 1945 to prevent further destruction following the Second World War, the United Nations (UN) Disarmament and International Security Committee was created as the first committee of the United Nations General Assembly. Also known as the First Committee, DISEC was established in 1946 as a result of the atomic bombs Japan faced in 1945, making its first agenda items focus on political and security issues. DISEC was set up to provide countries with a space to discuss issues related to peace and security.

As the UN Charter states, DISEC's role in the General Assembly is to help set "general principles of cooperation in the maintenance of international peace and security," including disarmament and arms control.

b. What Does DISEC Do?

DISEC focuses on the regulation of all member states' armaments, with particular care taken to work towards the UN's goal of total disarmament. It can make recommendations for UN member states or the Security Council to consider, since it cannot directly influence the Security Council's decisions. DISEC addresses topics centered on disarmament, global issues, and threats to peace that jeopardize international security. DISEC works closely with the UN Office for Disarmament Affairs (UNODA), which was created in 1998. UNODA focuses on all levels of disarmament, such as nuclear weapons and conventional weapons, and helps DISEC achieve its disarmament goals.

5. INTRODUCTION TO THE AGENDA ITEM

a. Key Terms and Concepts

According to the UN, chemical weapons are identified as together or separately;

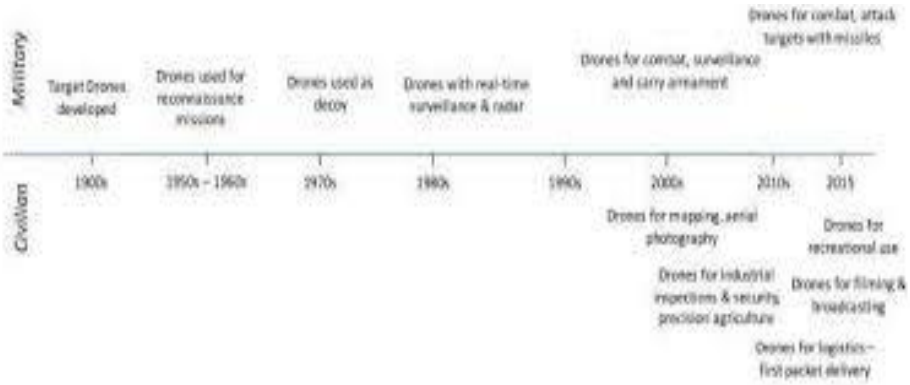
a) Toxic chemicals and their precursors, except where intended for purposes not prohibited under this convention, as long as the types and quantities are consistent with such purposes;
b) Munitions and devices, specifically designed to cause death or other harm through the toxic properties of those toxic chemicals specified in subparagraph (a), which would be released as a result of the employment of such munitions and devices;
c) Any equipment specifically designed for use directly in connection with the employment of munitions and devices specified in subparagraph (b).

Advanced technologies in wars provide the user with an enhanced level of support, thereby increasing effectiveness. Advanced technology weapons are more lethal to the enemy but safer for operators and non-combatants. They remain easy to operate, even under intense combat conditions.

Autonomous weapon systems are weapons that select and apply force to targets without human intervention. The most common types of weapons with autonomous functions are defensive systems. This includes systems such as antivehicle and antipersonnel mines, which, once activated, operate autonomously based on trigger mechanisms.

Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, have now become an essential tool for modern warfare, offering an edge on the battlefield. From the Russia-Ukraine conflict to the Iran-Israel tension and even in neighboring affairs between India and Pakistan, UAVs are altering the character of military confrontations all over the world.

The use of drones has changed the character of warfare with the zone in which ground forces are vulnerable to lethal attack extending to between six and nine miles behind the front lines. This has made trenches, fortified positions and armored vehicles much more vulnerable than they would have been previously.



Cyber warfare is no longer a theoretical future threat—it is an active, continuous conflict unfolding every day across global digital infrastructure. Unlike traditional warfare, this conflict has no clear battlefields, no uniforms, and often no immediate attribution. Power grids, healthcare systems, financial markets, universities, governments, and private enterprises have all become targets.

The emergence of artificial intelligence (AI) has accelerated this conflict dramatically. What we are witnessing today is not simply an increase in cybercrime, but the rise of a digital arms race, where nations, criminal syndicates, and non-state actors compete to gain strategic advantage through technology, automation, and data.

The integration of artificial intelligence in the domain of cyber warfare presents a paradoxical scenario. On the one hand, AI technologies provide cybersecurity professionals with advanced tools for threat detection, predictive analytics, and automated response mechanisms. These capabilities enable the development of more resilient defenses against sophisticated cyber threats.

On the other hand, the same AI technologies are being weaponized by adversaries to create more elusive and complex cyberattacks. AI-driven tactics include the creation of advanced malware that can

learn and adapt, conducting highly personalized social engineering attacks, and exploiting vulnerabilities with unprecedented speed and efficiency.

b. Current Use in Ongoing and Recent Conflicts

- Ukraine-Russia War

The Ukraine-Russia conflict has become a testing ground for innovation in drone warfare. Ukraine's June 2025 attack on Russian military bases destroyed or damaged 41 Russian aircraft, including advanced bombers, using drones smuggled into Russia on trucks and positioned near airfields. This operation, planned for over a year, demonstrates the strategic potential of asymmetric drone warfare.

Ukraine has developed AI-enabled autonomous navigation systems to counter Russian electronic warfare, with kamikaze drones destroying Russian tank columns despite jamming devices. The conflict has also seen the emergence of drone swarm technologies, representing the next evolution beyond the initial "one drone, one operator" concept.

- India-Pakistan Conflict

The May 2025 India-Pakistan conflict began after a terrorist attack in Pahalgam, Kashmir, that killed 26 tourists, escalating to air strikes where Indian armed forces hit terrorist targets in Pakistan. This crisis involved multidomain warfare including airstrikes, drones, cyberattacks, and naval maneuvers—representing an escalation under the nuclear umbrella.

- Israel-Iran Conflict

The current Israel-Iran conflict began with Israeli strikes targeting nuclear facilities, scientists, and military commanders, followed by Iranian retaliation with ballistic missiles and drones. Iran has reportedly launched over 100 drones toward Israel, while Israel has targeted civilian energy infrastructure, including the South Pars gas field, marking a shift from purely military to economic targets.

- Syrian Civil War

At the outbreak of the Syrian civil war in 2011, concerns were raised about both the security of Syria’s chemical weapon sites and the potential use of chemical weapons. In July 2012, a Syrian Foreign Ministry spokesman stated that no chemical or biological weapons will be used and that these types of weapons were in storage under security supervised by Syrian armed forces. They wouldn’t be used unless Syria was exposed to external aggression. In September 2012, the Syrian military began

moving chemical weapons from Damascus to the port of Tartus. It emerged that the Russian government helped set up between the US and Syria regarding the status of Syria's chemical weapons. They stated that the purpose of the movement of the chemical weapons was to secure them from opposing forces. In December 2012 various chemical weapons experts expressed skepticism that Syria was preparing to use chemical weapons, noting their "limited utility" in a civil war situation with fluid battle lines and Syria's comments that such use would be "suicide" in view of US threats of retaliation.

On the 21st of August 2013, the deadliest use of chemical weapons in the Syrian conflict happened in the Ghouta attack. Two areas in the suburbs around Damascus were struck by rockets that contained the chemical agent sarin. This attack resulted in an agreement to eliminate Syria's chemical weapons.

Following the Ghouta attack, the Khan Shaykun attack occurred in 2017 around Idlib. The town was struck by an airstrike by government forces; the release of a toxic gas, which included sarin or a similar substance, killed and injured lots of people. The OPCW-UN Joint Investigative Mechanism attributed the responsibility of the attack to the Syrian government, while the Syrian government said the attack was a "fabrication," and the Russian government said that the incident was staged.

One year later, the Douma chemical attack occurred. A gas was released, which allegedly included the sarin nerve agent, causing death and injuries. The attack was attributed to the Syrian Army by rebel forces in Douma and by the US, British, and French governments. A two-year-long investigation by the OPCW Investigation and Identification Team concluded that the Syrian Air Force perpetrated the attack using chlorine gas. On April 14, 2018, the US, France, and UK carried out a series of military strikes against two alleged chemical weapons facilities of the Syrian government.

d. Humanitarian and Environmental Impact

- Health Effects

Information about long-term health effects of chemical weapons is mainly anecdotal. Soldiers who have claimed that they were exposed to chemical warfare agents have often presented with unusual clinical syndromes or symptom complexes that have caused controversy.
Nerve agents, such as sarin, tabun, and soman, are believed to have the most severe long-term consequences. The effects of exposure to organophosphorus pesticides are very similar to those attributed to nerve agents. Animals and people who have developed acute symptoms as a result of exposure to nerve agents have been reported to have altered neuromuscular function, abnormal electroencephalograms, and impaired psychomotor performance that persists for at least 1 year after severe acute exposure. There is no evidence to suggest that nerve agents can cause cancer in humans. There are no particular groups of people who are more or less vulnerable to the toxic effects of nerve agents. There is limited data available on the reproductive and developmental effects of nerve agents. Therefore, it is not possible to draw any definitive conclusions. Effects on the unborn child are more likely to occur if the exposure to a nerve agent causes the mother to become unwell. Children would be expected to be affected by nerve agents in the same way as adults.

- Environmental Damage

Many conventional weapons have toxic constituents; others, such as depleted uranium, are also radioactive. Incendiary weapons such as white phosphorus are not only toxic but can also damage habitats through fire. While now restricted, the widespread use of chemical defoliants damaged public and ecological health across large areas of Vietnam.
Mustard gas and other agents in chemical weapons dumped after World War II are polluting the Baltic Sea's underwater sediment. Researchers found traces of arsenic-containing agents and mustard gas products in sediment and fish near three weapon disposal sites.

The warfare agents came from the German Army in WWII. After Germany's defeat, the Soviet Union dumped the captured chemical weapons in the Baltic Sea. The British and Americans did the same in the Skagerrak strait, between Norway and Sweden to the north and Denmark to the south. At the time, this disposal method was deemed less hazardous than burning the materials. The authors of the study say the Soviets jettisoned at least 50,000 metric tons (t) of German weapons into the Baltic Sea, some 15,000 t of which contained chemical agents. Previous reports concluded that most of these munitions probably had sunk into mud on the seafloor and were largely harmless. They frequently detected pollution some 250 m from the munition dump sites, but in some cases the contaminated area extended to 1 km, depending on currents at the sea bottom.

Adverse environmental impacts on human health include the generation of huge volumes of debris and waste and the release of hazardous materials such as asbestos, industrial chemicals, and fuels. Additionally, damage to industrial facilities can result in chemical spills and land and water contamination, while damage to water supply and wastewater sanitation facilities or the collapse of solid waste infrastructure can result in pollution.

Cityscapes reduced to rubble are emblematic of the use of EWIPA, but their cleanup is one of the most overlooked in recovery efforts.

In Syria, one-third of the housing stock has been damaged or destroyed, according to World Bank assessments. An estimated 15 million tons of debris were generated in Aleppo and 5.3 million tons in Homs.

In Iraq, around 63 cities and 1,556 villages were reportedly destroyed during the ISIL conflict between 2014 and 2017. National estimates indicate that over 55 million tons of debris were generated from residential buildings. This figure does not account for debris from destroyed government buildings or public infrastructure.

- Psychological Impacts

Attacks involving chemical weapons may induce significant mental and social effects in a number of ways—even when the agents induce low levels of mortality and physical morbidity. First, as the term 'bioterrorism' suggests, chemical attacks are associated with the experience of intense social and psychological distress, especially fear. Second, physical exposure to chemical agents may induce organic mental disorders (e.g., organic psychosis, delirium, dementia) (Benedek et al., 2002;

DiGiovanni, 1999). Third, exposure to any severe stressor—whether natural or human-made—is a risk factor for a range of long-term social and mental problems (including anxiety and mood disorders as well as non-pathological trauma and grief reactions) (Bromet & Havenaar, 2002). Fourth, fear of chemical attacks may be associated with epidemics of medically unexplained illness. Fifth, social problems may emerge after exposure to biological and chemical agents (e.g. population displacement; breakdown of community support systems; and social stigma associated with contagion or contamination).

By following up a random sample of 103 soldiers awarded a pension for gassing, it shows that they continued to experience a wide range of physical and psychological symptoms long after the respiratory or skin lesions had healed. Upper respiratory tract infections acquired in peacetime reinforced beliefs about the power of gas to cause serious and lasting harm, and a general debate flourished in the medical community about the capacity of chemical weapons to trigger latent tuberculosis.

These findings resonate with recent studies of US veterans who believe that they were gassed when engaged in the 1991 Gulf War. In a society that is increasingly 'chemophobic,' there is every reason to conclude that the psychological effects of chemical weapons are as potent today as when they were used during the First World War.

The 1980-1988 Iran-Iraq War inflicted enormous human costs, as each side sustained hundreds of thousands of casualties. According to Iranian government estimates, Iran sustained approximately 387 chemical attacks (via rocket, air, or artillery) during the 8-year war, which resulted in high rates of medical morbidity and psychological distress among an estimated 60,000 Iranian survivors.

Studies in war-afflicted civilians have reported high rates of war-related mental health problems. In a meta-analysis of 20 studies involving refugee populations living in high-income Western countries, Fazel et al. found that at least 9% of adult refugees were diagnosed with posttraumatic stress disorder (PTSD), and approximately 5% met criteria for major depression and generalized anxiety disorder. Prevalence estimates are much higher in civilians still living in war-afflicted countries. For example, as part of a national mental health survey in Afghanistan, Cardozo et al found very high prevalence rates for symptoms of anxiety (72%), depression (68%), and PTSD (42%). Similarly, among community samples involving survivors of war, conflict, or mass violence, de Jong et al. reported prevalence rates of PTSD symptoms of 37% in Algeria, 28% in Cambodia, 18% in Gaza, and 16% in Ethiopia.

Few studies have examined the mental health consequences of exposure to chemical weapon**s**. One study conducted in Japan 5 years after the 1995 sarin terrorist attack found that 17% of 34 participants met criteria for current PTSD. Studies among US soldiers who participated in secret military tests of mustard gas during World War II found that 50% experienced partial or full lifetime PTSD, and nearly one third met criteria for full current PTSD 50 years after the exposure.

e. Legal Framework and International Law

- Chemical Weapons Convention (CWC)

The Chemical Weapons Convention (CWC), officially the Convention on the Prohibition of the Development, Production, Stockpiling, and Use of Chemical Weapons and on their Destruction, is an arms control treaty administered by the Organisation for the Prohibition of Chemical Weapons (OPCW), an intergovernmental organization based in The Hague, Netherlands. CWC negotiations started in 1980 in the UN Conference on Disarmament. The convention opened for signature on January 13, 1993, and entered into force on April 29, 1997.

CWC is a multilateral treaty that bans chemical weapons and requires their destruction within a specified period of time. The treaty is of unlimited duration and is far more comprehensive than the 1925 Geneva Protocol, which outlaws the use but not the possession of chemical weapons.

The CWC is implemented by the Organization for the Prohibition of Chemical Weapons (OPCW), which is headquartered in The Hague with about 500 employees. The OPCW receives states-parties' declarations detailing chemical weapons-related activities or materials and relevant industrial activities. After receiving declarations, the OPCW inspects and monitors states-parties' facilities and activities that are relevant to the convention to ensure compliance.

The Chemical Weapons Convention prohibits:

- Developing, producing, acquiring, stockpiling, or retaining chemical weapons.
- The direct or indirect transfer of chemical weapons.
- Chemical weapons use or military preparation for use.
- Assisting, encouraging, or inducing other states to engage in CWC-prohibited activity.
- The use of riot control agents "as a method of warfare."

The CWC requires states-parties to declare in writing to the OPCW their chemical weapons stockpiles, chemical weapons production facilities (CWPFs), relevant chemical industry facilities, and other weapons-related information. This must be done within 30 days of the convention's entry into force for each member state.

Chemical Weapons Stockpiles—States-parties must declare all chemical weapons stockpiles, which are broken down into three categories:

- Category 1: chemical weapons based on Schedule 1 chemicals, including VX and sarin. (See below for an explanation of "scheduled" chemicals.)
- Category 2: chemical weapons based on non-Schedule 1 chemicals, such as phosgene.
- Category 3: chemical weapons, including unfilled munitions, devices, and equipment designed specifically to employ chemical weapons.

Other weapons-related declarations states parties must make include:

- Chemical weapons production facilities on their territories since January 1, 1946.
- Facilities (such as laboratories and test sites) designed, constructed, or used primarily for chemical weapons development since January 1, 1946.
- "Old" chemical weapons on their territories (chemical weapons manufactured before 1925 or those produced between 1925 and 1946 that have deteriorated to such an extent that they are no longer usable).
- "Abandoned" chemical weapons (abandoned by another state without consent on or after January 1, 1925).

- Plans for destroying weapons and facilities.
- All transfers or receipts of chemical weapons or chemical weapons production equipment since January 1, 1946.
- All riot control agents are in their possession.

Chemical Industry—The CWC requires states-parties to declare chemical industry facilities that produce or use chemicals of concern to the convention. These chemicals are grouped into "schedules," based on the risk they pose to the convention. A facility producing a Schedule 1 chemical is considered a Schedule 1 facility.

- Schedule 1 chemicals and precursors pose a "high risk" to the convention and are rarely used for peaceful purposes. States-parties may not retain these chemicals except in small quantities for research, medical, pharmaceutical, or defensive use. Many Schedule 1 chemicals have been stockpiled as chemical weapons.
- Schedule 2 chemicals are toxic chemicals that pose a "significant risk" to the convention and are precursors to the production of Schedule 1 or Schedule 2 chemicals. These chemicals are not produced in large quantities for commercial or other peaceful purposes.
- Schedule 3 chemicals are usually produced in large quantities for purposes not prohibited by the CWC but still pose a risk to the convention. Some of these chemicals have been stockpiled as chemical weapons.

The CWC also requires the declaration of facilities that produce certain nonscheduled chemicals.

- The Geneva Conventions

The 1949 Geneva Conventions and their Additional Protocols are international treaties that contain the most important rules limiting the barbarity of war. They protect people who do not take part in the fighting (civilians, medics, aid workers) and those who can no longer fight (wounded, sick, and shipwrecked troops, prisoners of war).

The Geneva Conventions define the basic rights and protections afforded to those non-combatants who fulfill the criteria of being protected persons. The Geneva Conventions concern only protected non-combatants in war. The Geneva Conventions define the rights of civilians, prisoners of war, and military personnel; establish protections for the wounded, sick, and shipwrecked; and provide humanitarian protections for civilians in and around war zones. The use of conventional weapons in wartime is addressed by the Hague Conventions of 1899 and 1907 and the 1980 Convention on Certain Conventional Weapons, while the use of biological and chemical weapons in international armed conflicts is addressed by the 1925 Geneva Protocol.

Violations of these earlier conventions constituted some of the grounds for the judgments of the Nuremberg, Tokyo, and other war crimes tribunals. For grave breaches of the Geneva Conventions, states are obligated to establish and exercise universal jurisdiction. All four of the 1949 Geneva Conventions—in Articles 49, 50, 129, and 146 (respectively)—contain provisions obligating parties to search for alleged violators, regardless of their nationality or place of offense, and to either try violators before their own courts or to hand them over to another party for trial. This principle of universal jurisdiction is rooted in the belief that grave breaches of the Geneva Convention are such serious crimes that all states have an obligation to bring those responsible to justice.

The ICRC has produced commentaries on each of the Geneva Conventions and their Additional Protocols to promote better understanding of and respect for these vital treaties. They are currently being updated to incorporate developments in the application and interpretation of these treaties since their negotiation. The main aim of the updated Commentaries is to give people an understanding of the law as it is currently interpreted so that it can be applied effectively in today's armed conflicts. They are an essential tool for practitioners, reaffirming the continued relevance of the Conventions, generating respect for them, and strengthening protection for people affected by armed conflict.

- The First Geneva Convention

The Geneva Convention for the Amelioration of the Condition of the Wounded and Sick in Armed Forces in the Field of August 12, 1949. The First Geneva Convention protects soldiers who are hors de combat (out of the battle). The 10 articles of the original 1864 version of the Convention have been expanded in the First Geneva Convention of 1949 to 64 articles that protect the following:

• Wounded and sick soldiers
• Medical personnel, facilities and equipment
• Wounded and sick civilian support personnel accompanying the armed forces
• Military chaplains
• Civilians who spontaneously take up arms to repel an invasion.

- The Second Geneva Convention

The Geneva Convention for the Amelioration of the Condition of Wounded, Sick and Shipwrecked Members of Armed Forces at Sea of August 12, 1949 The Second Geneva Convention adapts the protections of the First Geneva Convention to reflect conditions at sea. It protects wounded and sick combatants while on board ship or at sea. Its 63 articles apply to the following:

• Armed forces members who are wounded, sick or shipwrecked
• Hospital ships and medical personnel
• Civilians who accompany the armed forces

The Fourth Geneva Convention: The Geneva Convention Relative to the Protection of Civilian Persons in Time of War of August 12, 1949. Civilians in areas of armed conflict and occupied territories are protected by the 159 articles of the Fourth Geneva Convention.

- The Third Geneva Convention

The Geneva Convention Relative to the Treatment of Prisoners of War of August 12, 1949 The Third Geneva Convention sets out specific rules for the treatment of prisoners of war (POWs). The Convention’s 143 articles require that POWs be treated humanely, adequately housed, and receive sufficient food, clothing, and medical care. Its provisions also establish guidelines on labor, discipline, recreation, and criminal trial. Note that prisoners of war may include the following:

• Members of the armed forces
• Volunteer militia, including resistance movements
• Civilians accompanying the armed forces.

- The Fourth Geneva Convention

The Geneva Convention Relative to the Protection of Civilian Persons in Time of War of August 12, 1949: Civilians in areas of armed conflict and occupied territories are protected by the 159 articles of the Fourth Geneva Convention.

- Article 35 of the Geneva Convention states that the use of weapons that "cause superfluous injury or unnecessary suffering," as well as means of warfare that "cause widespread, long-term, and severe damage to the natural environment," is prohibited.

- Art. 51 and 54 outlaw indiscriminate attacks on civilian populations and destruction of food, water, and other materials needed for survival.

- Convention on Certain Conventional Weapons (CCW)

The Convention on Prohibitions or Restrictions on the Use of Certain Conventional Weapons Which May Be Deemed to Be Excessively Injurious or to Have Indiscriminate Effects as amended on 21 December 2001, usually referred to as the Convention on Certain Conventional Weapons (CCW), is a key international humanitarian law instrument. Its purpose is to ban or restrict the use of specific types of weapons that are considered to cause unnecessary or unjustifiable suffering to combatants or to affect civilians indiscriminately.

The aim of the convention and its protocols is to provide new rules for the protection of civilians from injury by weapons that are used in armed conflicts and also to protect combatants from unnecessary suffering. The convention covers fragments that are undetectable in the human body by X-rays, landmines and booby traps, incendiary weapons, blinding laser weapons, and the clearance of explosive remnants of war. Parties to the convention must take legislative and other actions to ensure compliance with the convention.

CCWC includes five protocols dealing with specific weapons. The convention and its annexed protocols apply in all types of armed conflict, both international and non-international. This was not the case when the convention was first adopted, but the scope was expanded by two conferences in 1996 and 2001. Some provisions also apply after open hostilities have ended, such as the rules in protocols II and V about minimizing the dangers from mines and other ordnance.

- CCW Protocols

1/ Protocol I – Non-Detectable Fragments
Prohibits the use of any weapon designed to injure by fragments that cannot be detected in the human body by X-rays.

2/ Amended Protocol II – Mines, Booby Traps and Other Devices, as amended on 3 May 1996
Prohibits the use of nondetectable antipersonnel mines and their transfer, and prohibits the use of non-self-destructing and non-self-deactivating mines outside fenced, monitored, and marked areas. Seeks to limit the indiscriminate damage caused by landmines and requires High Contracting Parties to take all feasible precautions to protect civilians when using these weapons. Amended Protocol II is the only legally binding instrument that covers Improvised Explosive Devices (IED). A limited number of State Parties are still party to Original Protocol II.

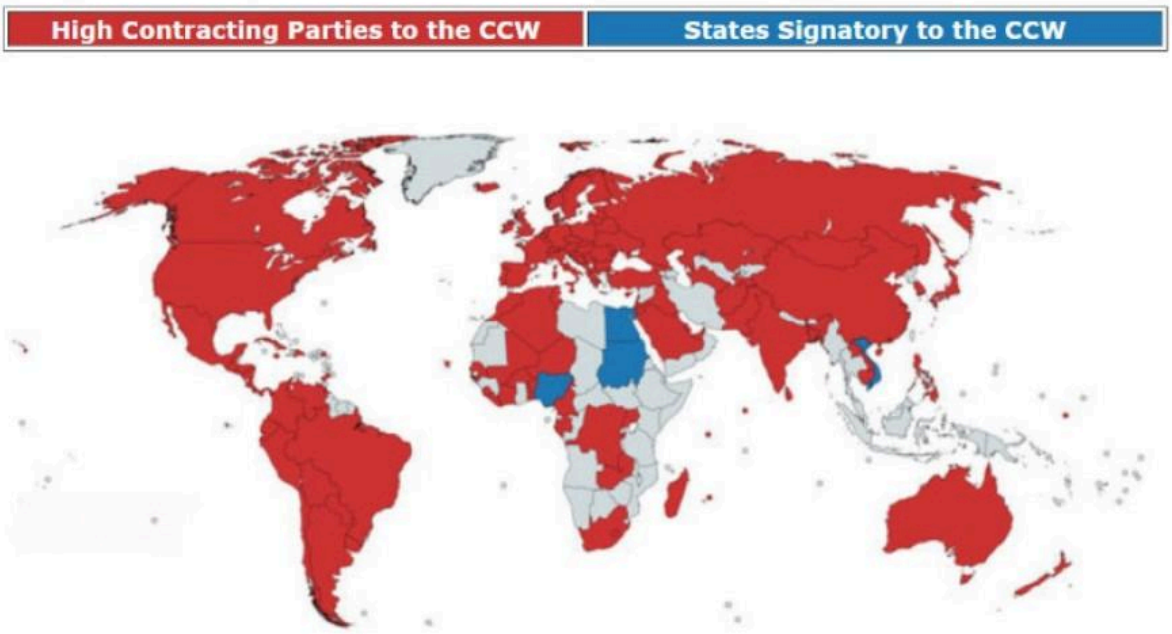
3/ Protocol III – Incendiary Weapons
Prohibits the use of weapons primarily designed to set fire to objects or cause burn injuries against civilians

.

4/ Protocol IV – Blinding Laser Weapons
Prohibits the use and transfer of laser weapons designed to cause permanent blindness.
5/ Protocol V – Explosive Remnants of War
Prevents and minimizes the humanitarian impact of unexploded ordnance and abandoned explosive weapons. Includes provisions on clearance and destruction of ERW, measures for the protection of civilians, recording the use of explosive ordnance, international cooperation and assistance, and victim assistance.

By ratifying the CCW, states commit to the:

- Prohibition of the use of any weapon of which the primary effect is to injure by fragments that are not detectable in the human body by X-rays;
- Prohibition and regulation of the use and transfer of non-detectable anti-personnel mines, boobytraps, and other devices;
- Prohibition of, in all circumstances, making civilians the object of attack by incendiary weapons;
- Prohibition of the use of laser weapons specifically designed to cause permanent blindness and the transfer of such weapons to any State or non-State entity;
- Requirement of Parties to a conflict to take measures to reduce dangers posed by explosive remnants of war.



6. Previous UN Actions and Resolutions

UNSC Resolution 1540

United Nations Security Council resolution 1540 was adopted unanimously on 28 April 2004 regarding the non-proliferation of weapons of mass destruction.

The resolution establishes the obligations under Chapter VII of the United Nations Charter for all member states to develop and enforce appropriate legal and regulatory measures against the proliferation of chemical, biological, radiological, and nuclear weapons and their means of delivery, in particular, to prevent the spread of weapons of mass destruction to non-state actors.

The cooperation encouraged by the CWC at the State level aligns with the collective global efforts mandated by UNSCR 1540, emphasizing the importance of international cooperation in preventing the proliferation of WMDs and safeguarding global security.

UNSC Resolution 2209

The United Nations Security Council Resolution 2209 condemns any use of chemicals as a weapon in the Syrian Civil War and threatens to use force if chemical weapons are used again in the conflict.

The Council expressed support for the 4 February OPCW Executive Council decision to continue the work of the fact-finding mission, in particular to study all available information relating to allegations of chemical weapons use in Syria. Stressing that those responsible for use of chemical weapons —

including chlorine — must be held accountable, the Council called on all parties to fully cooperate with the mission.

UNSC Resolution 2218

United Nations Security Council Resolution 2118 (2013) was adopted unanimously on 27 September 2013. It was passed in response to the chemical weapons attack in Ghouta, Syria, in August 2013, which caused hundreds of civilian deaths and raised serious concerns about the use of weapons of mass destruction during the Syrian civil war.

The resolution provides that:

- Syria is prohibited from developing, producing, acquiring, stockpiling, or using chemical weapons.
- Syria must join and comply with the Chemical Weapons Convention (CWC).
- All chemical weapons, related materials, and production facilities must be declared and destroyed under international supervision.
- The destruction process is carried out jointly by the UN and the Organisation for the Prohibition of Chemical Weapons (OPCW).
- In case of non-compliance, the Security Council may impose measures under Chapter VII of the UN Charter.

OPCW Fact-Finding Mission (FFM)

With repeated allegations of the use of toxic chemicals for hostile purposes at a number of locations in the Syrian Arab Republic, on 29 April 2014, the Director-General announced the formation of an FFM. The establishment of the FFM was based on the general authority of the OPCW Director-General to seek to uphold at all times the object and purpose of the Chemical Weapons Convention. This authority is reinforced by the relevant decisions of the OPCW Executive Council and the United Nations Security Council resolution (UNSCR) 2118 (2013); and its acceptance by the Syrian Arab Republic.

The OPCW Fact-Finding Mission (FFM) is responsible for determining whether toxic chemicals have been used as weapons in the Syrian Arab Republic. The mandate does not include identifying who is responsible for any alleged attacks.

The Mission took over the work of the OPCW-UN Joint Mission in Syria, which had been formed to oversee the elimination of the Syrian chemical weapons program, and which ended its activities on 30 September 2014.On 4 September 2014, the head of the Joint Mission reported to the UN Security Council that 96% of Syria's declared stockpile, including the most dangerous chemicals, had been destroyed and preparations were underway to destroy the remaining 12 production facilities, a task to be completed by the OPCW Mission. On 4 January 2015, the OPCW stated that destruction was

completed, though since then previously undeclared traces of compounds in a Syrian government military research site have been reported.

7. Key Challenges

Lack of Enforcement Mechanisms

By 2010, the establishment and entry into force of the Chemical Weapons Convention (CWC), the declaration and elimination of most known CW programs, and a sustained period of non-use of chemical weapons had driven concern about chemical weapons to a few esoteric corners. 1 As the number of states possessing chemical weapons steadily declined, the CWC appeared to be in the final technical stage of implementation, and some countries began questioning the sustainability of the Organization for the Prohibition of Chemical Weapons (OPCW), the implementing body of the treaty. It seemed that chemical weapons had been successfully managed and controlled through a robust international system of treaties, laws, and cooperative arrangements that led to a near-universal CW disarmament. For many, it was time to "declare victory" on chemical weapons and focus on other more pressing threats—nuclear, biological, and even cyber.

Unfortunately, this optimistic view has proven unwarranted. Syrian use of chemical weapons persisted even as the Assad regime joined the CWC and the international community embarked on a large-scale effort to remove and destroy the country's declared CW stockpiles and facilities. Since 2015, the numbers and types of CW attacks have increased, with at least 336 documented cases in Syria alone. 2 While the overwhelming preponderance of uses have occurred throughout the Syrian conflict, CW attacks have occurred in three other countries—Iraq, Malaysia, and the United Kingdom—for a variety of purposes and involving a variety of agents.

The prospect of large-scale, state-on-state chemical warfare has continued to fade, but states have pursued and employed chemical weapons for a variety of tactics and strategies that include counterinsurgency, assassination, coercion, terror, collective punishment, and signaling. Even more alarming, such use has not been confined to states operating outside of the CWC. Rather, state parties to the CWC, specifically Syria and Russia, have used and enabled the use of chemical weapons despite their status as treaty adherents. Small-scale, high-impact use of chemical weapons has been used to suppress internal conflicts, intimidate or eliminate political adversaries, and engage in grey zone conflict in ways that challenge the ability of countries to respond effectively.

Additionally, non-state actors have developed and used "military agents" (specifically sulfur mustard, though it is unlikely to be as pure as traditional military-grade sulfur mustard agents) and chlorine with a range of improvised delivery systems. ISIL demonstrated that access to territorial safe havens, facilities, and personnel could be leveraged into more advanced chemical agent and delivery capabilities, which the group used in targeted regional military or insurgent operations rather than in classic terrorist attacks.

CW users have also employed significant improvisation—agents, munitions, delivery systems, and concepts of operation—even when more traditional options are available. Such improvisation—from chlorine barrel bombs to perfume bottles filled with thousands of lethal doses of advanced military

nerve agent—is a hallmark of the evolving utility and application of chemical weapons to contemporary conflicts. Meanwhile, international organizations such as the United Nations and the OPCW have shown little ability to enforce obligations or hold perpetrators accountable. Moreover, the little progress made has required a shift to more contentious forms of diplomacy and resulted in the collapse of consensus within the OPCW Executive Council. Finally, chemical agents that fall outside of the traditional schedule of military agents—such as chlorine and other TICs, fentanyls and other incapacitants and opioids, and Novichoks—are demonstrating growing potential for use. Meanwhile, advances in chemical engineering and rapidly expanding global supply networks are making weaponized chemicals increasingly accessible.

Potential users and proliferators of chemical weapons can also be restrained through a perception that these actions fundamentally lack benefit or utility in terms of meeting desired objectives. If an actor fundamentally believes chemical weapons have no utility or functionality, then the motivation for their use is low, and restraint is a natural outcome. Furthermore, enhanced CW defenses or protection offer additional ways of reinforcing perceived lack of benefit, leading to what some call deterrence by denial.

However, it is impossible to deny the benefits of CW use if their utility to the adversary is fundamentally misunderstood. Projected bias and unchallenged assumptions about the utility of chemical weapons have led to a serious underestimation of their utility for counterinsurgency, special operations, intimidation, coercive grey zone or hybrid tactics, and large-scale civilian terror operations. This faulty analysis has been fostered by a tendency to evaluate the efficacy of chemical use solely in terms of direct lethality or tactical impact rather than as a psychological weapon of terror or intimidation. Additionally, in the case of Syria, the role chemical weapons play in combined chemical and conventional military operations has been underappreciated. Discounting the “logic” of CW use based on misperceptions of benefit to the user can also aid proponents of false flag conspiracy theories and other disinformation tactics.

In addition to new technology, there are new and emerging agents, as well as old agents being used in new ways. Many such chemical agents—chlorine and other TICs, fentanyls and other deadly pharmaceutically-based agents (PBAs), and Novichoks—are not fully included on the CWC schedules and can be transferred and used in ways that challenge or elude traditional verification and controls. For example, chlorine is too ubiquitous to control, and a significant number of actual and potential CW agents, like fentanyls, lie outside of the CWC schedules. Following the Skripal attack, some Novichok formulations were added to Schedule 1, but it is not clear if and how precursors to these agents can be regulated. In addition, PBAs, such as fentanyls, are a growing concern because they are easily produced, acquired, and weaponized and can be highly lethal. Fentanyl is 50 times more potent than heroin and, in drug-trafficking operations, is frequently mixed with heroin and cocaine or made into counterfeit pills. Fentanyl seizures have made headlines in recent years, with millions of lethal doses being confiscated in single drug busts. In August 2019, a three-state drug bust seized 30 kilograms of fentanyl—enough to kill roughly 14 million people.

Difficulty in Monitoring Advanced Technologies

The use of facial recognition technology, notably from Clearview AI, in the war has been transformative but also raises serious ethical concerns. While it aids in identifying Russian soldiers

and missing persons, it also has potential for mass surveillance, misidentification, and privacy violations. Its unregulated use threatens fundamental rights, and Clearview AI has faced legal actions in different countries for breaching data-protection laws. These concerns are heightened by the technology's possible continued use after the war.
Russia's use of AI in information warfare has intensified the conflict, with AI-generated disinformation, deepfakes, and voice cloning spreading false narratives and destabilizing Ukrainian society. Deepfake videos, such as those depicting Ukrainian leaders surrendering, are used to erode trust in media and pose significant threats to social cohesion and democracy.

As Ukraine increasingly relies on these technologies in the war, the need to withhold sensitive information and to maintain public trust conflicts. Striking a balance between national security and human rights is essential. International law allows for limited derogations from human rights during emergencies, but these must adhere to the principles of necessity and proportionality.

These crucial technologies require stringent safeguards to protect democratic values and human rights in Ukraine. Key recommendations to do so include regulating remote-sensing technologies; harmonizing the country's AI and data-protection laws with EU standards; adopting impact assessment frameworks; creating "regulatory sandboxes" for the safe development, testing, and adaptation of AI innovations; and enhancing media literacy programs to counter AI-generated disinformation. Through these measures, Ukraine can harness advanced technologies for defense and reconstruction while ensuring that wartime measures do not undermine its democratic future.

As in Ukraine, emerging technologies have played a critical role on the battlefield in Gaza and are reshaping the conduct of the war. However, the Gaza war has yielded fewer insights on the impact of emerging technologies, as Israel was already a global leader in incorporating technological solutions. Further, the conflict was less evenly matched—Israel's quick bursts of activity and Hamas's lack of technological prowess have not resulted in the leapfrogging technological achievements that have featured in Ukraine.

Among the technologies that have been publicly acknowledged, the most notable is the AI-enabled decision-support systems (AI-DSS) that the Israel Defense Force (IDF) uses for targeting, which has dramatically accelerated the processing and analysis of battlefield information. AI tools have also supported Israeli forces in tracking the movements of suspected Hamas operatives at checkpoints across Gaza and the West Bank. To support these systems, Israel has significantly increased its demand for data storage and cloud computing, drawing heavily on commercial providers as well as its startup ecosystem to rapidly field new technology for the battlefield.

The conflicts in Ukraine and the Middle East have shown how emerging technologies—particularly unmanned vehicles, AI, and information warfare—are reshaping combat, forcing militaries to adapt or risk obsolescence. Modern conflicts are increasingly defined by speed, adaptability, and innovation. Ultimately, the next generation of warfare will not be defined solely by who possesses the most advanced technology, but by who can integrate, adapt, and counter it the fastest.

Responsibility for Autonomous Weapons

It is undisputed that humans must retain responsibility for the development and use of autonomous weapon systems (AWS) because machines cannot be held accountable for violations of international

humanitarian law (IHL). However, the critical question of how, in practice, humans would be held responsible for IHL violations involving AWS has not featured strongly in the policy debate on AWS. This report aims to offer a comprehensive analysis of that very question.

Any weapon system with autonomy in its critical functions—that is, a weapon system that can select (search for, detect, identify, track, or select) and attack (use force against, neutralize, damage, or destroy) targets without human intervention.

After the initial launch or activation by a human operator, it is the weapon system itself—using its sensors, computer programming (software), and weaponry—that takes on the targeting functions that would otherwise be controlled by humans. This working definition encompasses any weapon system that can independently select and attack targets, including some existing weapons and potential future systems. The definition provides a useful basis for a legal analysis by delineating the broad scope of the discussion about autonomous weapon systems without the need to immediately identify the systems that raise legal concerns. In that sense, the definition is not intended to prejudge the level of autonomy in weapon systems that may, or may not, be considered lawful. Rather, the ICRC has proposed that states determine where these limits must be placed by assessing the type and degree of human control required in the use of weapon systems to carry out attacks—at a minimum, for compliance with IHL and, in addition, to satisfy ethical considerations.

Autonomous weapon systems, as defined, are not specifically regulated by IHL treaties. However, it is undisputed that any autonomous weapon system must be capable of being used, and must be used, in accordance with IHL. The responsibility for ensuring this rests, first and foremost, with each state that is developing, deploying, and using weapons.

While the primary subjects of IHL are the parties to an armed conflict, the rules on the conduct of hostilities—notably the rules of distinction, proportionality, and precautions in attack—are addressed to those who plan, decide upon, and carry out an attack.

The core legal obligations for a commander or operator in the use of weapon systems include the following: to ensure distinction between military objectives and civilian objects, combatants and civilians, and active combatants and those hors de combat; to determine whether the attack may be expected to cause incidental civilian casualties and damage to civilian objects, or a combination thereof, which would be excessive in relation to the concrete and direct military advantage anticipated, as required by the rule of proportionality; and to cancel or suspend an attack if it becomes apparent that the target is not a military objective or is subject to special protection, or that the attack may be expected to violate the rule of proportionality, as required by the rules on precautions in attack.

These IHL rules create obligations for human combatants in the use of weapons to carry out attacks, and it is combatants who are both responsible for respecting these rules and who will be held accountable for any violations. As for all obligations under international law, these legal obligations and accountability for them cannot be transferred to a machine, computer program, or weapon system.

Those who plan, decide upon, and carry out an attack using an autonomous weapon system must, therefore, ensure that the weapon system and the way it is used preserve their ability to make these necessary legal judgments and thereby ensure compliance with IHL. It follows that an autonomous weapon system will raise concerns under IHL if—through its design, performance, and/or method of

use—it impedes commanders or operators in making these legal judgments. For example, if a mobile autonomous weapon system searches for targets over a wide area and for a long duration, without human supervision and communication, the commander who authorized the launch of the weapon and the operator who activated it will not know exactly where and when an attack will take place. This raises questions of whether they will be able to ensure distinction, judge proportionality, or take precautions should the circumstances change.

The absence of precise international obligations of transparency and information on how states conduct their military operations is already a matter of serious concern for those who would like to ensure that violations of the applicable rules can be investigated. Adding to the 'black box' resulting from the lack of transparency and information on the conduct of hostilities, the 'black box' of systems enabled by strong artificial intelligence would contribute to the further weakening of such possibilities. States that use intelligent AWS could argue that prohibited military attacks are the result of an operation by intelligent software, which cannot be explained or predicted by the user and therefore cannot be attributed to them. Even the hypothesis of such a possibility is unacceptable.

8. Questions a Resolution Paper Must Address

1. How can affected countries prevent the ongoing environmental and civilian damage caused by chemical weapons that have been used in the past conflicts and attacks?

2. What mechanisms can be established and forced by the international community to monitor and regulate the possession and development of chemical weapons for state and non-state actors transparently under current international law?

3. How can strict regulations be enforced if such weapons are not fully prohibited?

4. How can unintended harm caused by AI decision-making be prevented?

5. How does the involvement of non-state actors complicate efforts to control advanced weapon technologies? And how can it be prevented?

6. What role should international organizations play in addressing the ethical and legal challenges posed by autonomous weapons and emerging military technologies, and how effective can they be in ensuring global cooperation and compliance?

7. How do black markets and the dual-use nature of emerging technologies complicate international regulation, and what measures can be taken to prevent civilian technologies from being repurposed for military or illegal use?

8. What policy approaches and international solutions can be implemented to address the risks of autonomous weapons and chemical technologies, while balancing technological progress, national security, and humanitarian concerns?

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